

Text as Data HW 1

Clarke Campbell

Text as Data: Homework 1

Clarke Campbell

September 11, 2026

Pre-Existing Tutorial Information

```
# In these code chunks, I am loading and setting up the data  
# like it was done in the tutorial.  
library(readr)  
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(tidyr)
library(stringr)
library(tidytext)
library(ggplot2)
library(forcats)
library(tibble)
library(scales)
```

Attaching package: 'scales'

The following object is masked from 'package:readr':

```
col_factor
```

```
file_a <- "texts/A07594__Circle_of_Commerce.txt"
file_b <- "texts/B14801__Free_Trade.txt"
```

```
# Read the raw text files into R
```

```
text_a <- read_file(file_a)
```

```
text_b <- read_file(file_b)
```

```
# Combine into a tibble for tidytext workflows
```

```
texts <- tibble(
  doc_title = c("Text A", "Text B"),
  text = c(text_a, text_b)
)
```

```
texts
```

```
# A tibble: 2 x 2
```

```
  doc_title text
```

```
  <chr>      <chr>
```

```
1 Text A    THE CIRCLE OF COMMERCE. The Prooeme. HERODOTVS in his CLIO, reporte~
```

```
2 Text B    CAP. I. The Causes of the want of Money in England. IT hauing pleas~
```

```
data("stop_words")
```

```
custom_stopwords <- tibble(
```

```
  word = c(
```

```

    "vnto", "haue", "doo", "hath", "bee", "ye", "thee"
  )
)

all_stopwords <- bind_rows(stop_words, custom_stopwords) %>%
  distinct(word)

```

Section I

```

words <- texts |>
  unnest_tokens(word, text)

# Here, I am removing the stopwords from my broader 'words' tibble.
words_stopwords_removed <- words |>
  mutate(word = str_to_lower(word)) |>
  anti_join(all_stopwords, by = "word")

# Here, I am counting the words in my broader 'words' tibble.
total_word_count <- words |>
  count(doc_title)

# Here, I am counting the words from my stopword-removed
# 'words_stopwords_removed' tibble, joining the counts of
# both documents, renaming and relocating the columns
# appropriately, and creating columns for the removed tokens'
# number and percentage.
corpus_diagnostics <- words_stopwords_removed |>
  count(doc_title) |>
  full_join(total_word_count, by = join_by(doc_title)) |>
  rename(tokens_after_stopwords = n.x, total_tokens = n.y) |>
  relocate(tokens_after_stopwords, .after = total_tokens) |>
  group_by(doc_title) |>
  mutate(
    tokens_removed = (total_tokens - tokens_after_stopwords),
    percent_removed = (tokens_removed / total_tokens * 100)
  ) |>
  ungroup()

corpus_diagnostics

```

```
# A tibble: 2 x 5
  doc_title total_tokens tokens_after_stopwords tokens_removed percent_removed
  <chr>      <int>          <int>          <int>          <dbl>
1 Text A      33889          14694          19195          56.6
2 Text B      19922           8004          11918          59.8
```

Narrative:

56.64% of Document A's words were removed and 59.82% of Document B's words were removed by excluding the stopwords from each of them. With a difference of only ~3.18%, I would say these percentages are approximately the same. The large magnitude of these percentages indicates that a very substantial portion of both texts has been removed by excluding stopwords. While we are starting with the assumption that stopwords provide little-to-no meaning and will likely distract from our textual analysis, we should be careful to ensure that, given the substantial reduction in words across both texts, removing them does not greatly alter the meaning and outcomes of what we are assessing.

Section II

```
# Here, I am counting the frequency of words
# across both documents before stopwords removal
# and sorting them by the most frequent words.

word_frequency_no_removals <- words |>
  count(doc_title, word, sort = TRUE) |>
  pivot_wider(
    names_from = doc_title,
    values_from = n,
    values_fill = 0
  ) |>
  mutate(max_n = pmax(`Text A`, `Text B`)) |>
  arrange(desc(max_n))

word_frequency_no_removals
```

```
# A tibble: 6,829 x 4
  word `Text A` `Text B` max_n
  <chr>   <int>   <int> <int>
1 the     2375    1598  2375
2 of      1612    1142  1612
3 and     1170     763  1170
```

```

4 in      870      559  870
5 to      818      503  818
6 is      586      296  586
7 that    519      314  519
8 a       484      263  484
9 for     381      226  381
10 his    348      142  348
# i 6,819 more rows

```

#These are the 15 highest-frequency words overall:

```

word_frequency_no_removals |>
  slice_head(n = 15)

```

A tibble: 15 x 4

```

  word   `Text A` `Text B` max_n
  <chr>   <int>   <int> <int>
1 the     2375    1598  2375
2 of      1612    1142  1612
3 and     1170     763  1170
4 in       870     559   870
5 to       818     503   818
6 is       586     296   586
7 that    519     314   519
8 a        484     263   484
9 for      381     226   381
10 his     348     142   348
11 or      342     172   342
12 it      321     196   321
13 as      318     169   318
14 this    259     172   259
15 which   243     190   243

```

#Next, I am plotting these in ggplot.

```

word_frequency_no_removals |>
  slice_head(n = 15) |>
  pivot_longer(
    cols = c(`Text A`, `Text B`),
    names_to = "doc_title",
    values_to = "n"
  ) |>
  mutate(word = fct_reorder(word, n, .fun = max)) |>

```

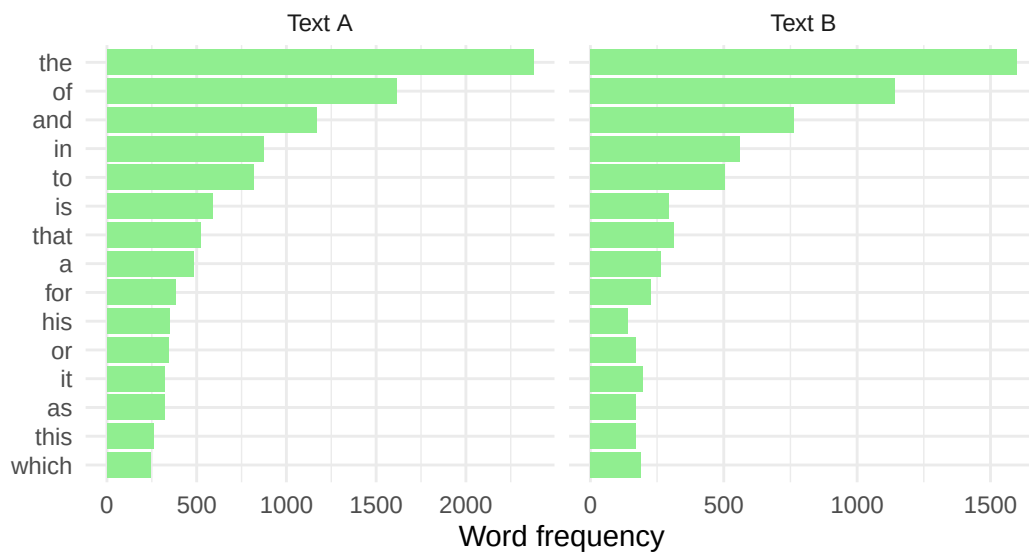
```

ggplot(aes(n, word)) +
  geom_col(fill = "lightgreen") +
  facet_wrap(~ doc_title, scales = "free_x") +
  labs(
    title = "Most frequent words (including all stopwords)",
    subtitle = paste0(
      "Top 15 words by maximum frequency across both texts"
    ),
    x = "Word frequency",
    y = NULL
  ) +
  theme_minimal()

```

Most frequent words (including all stopwords)

Top 15 words by maximum frequency across both texts



```

# Here, I am counting the frequency of words
# across both documents *AFTER* stopwords removal
# and sorting them by the most frequent words.

word_frequency_stopwords_removed <- words_stopwords_removed |>
  count(doc_title, word, sort = TRUE) |>
  pivot_wider(
    names_from = doc_title,
    values_from = n,

```

```

    values_fill = 0
  ) |>
  mutate(max_n = pmax(`Text A`, `Text B`)) |>
  arrange(desc(max_n))

```

word_frequency_stopwords_removed

```

# A tibble: 6,434 x 4
  word      `Text A` `Text B` max_n
  <chr>      <int>   <int> <int>
1 trade      232     185   232
2 exchange   186        4   186
3 fo         184        0   184
4 malynes    158        0   158
5 merchants  126     42   126
6 mony       118        8   118
7 hee        115     16   115
8 kingdome   113     75   113
9 commodities 96     18    96
10 money      84     81    84
# i 6,424 more rows

```

#These are the 15 highest-frequency words overall:

```

word_frequency_stopwords_removed |>
  slice_head(n = 15)

```

```

# A tibble: 15 x 4
  word      `Text A` `Text B` max_n
  <chr>      <int>   <int> <int>
1 trade      232     185   232
2 exchange   186        4   186
3 fo         184        0   184
4 malynes    158        0   158
5 merchants  126     42   126
6 mony       118        8   118
7 hee        115     16   115
8 kingdome   113     75   113
9 commodities 96     18    96
10 money      84     81    84
11 alfo       78        0    78
12 fuch       69        0    69

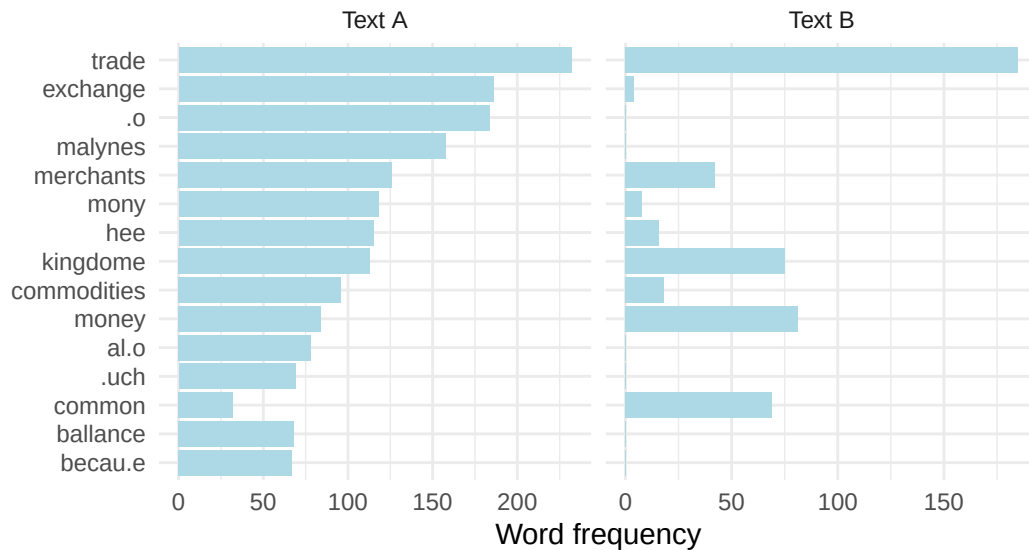
```

13 common	32	69	69
14 ballance	68	0	68
15 becaufe	67	0	67

#Next, I am plotting these in ggplot.

```
word_frequency_stopwords_removed |>
  slice_head(n = 15) |>
  pivot_longer(
    cols = c(`Text A`, `Text B`),
    names_to = "doc_title",
    values_to = "n"
  ) |>
  mutate(word = fct_reorder(word, n, .fun = max)) |>
  ggplot(aes(n, word)) +
  geom_col(fill = "lightblue") +
  facet_wrap(~ doc_title, scales = "free_x") +
  labs(
    title = "Most frequent words (stopwords removed)",
    subtitle = paste0(
      "Top 15 words by maximum frequency across both texts"
    ),
    x = "Word frequency",
    y = NULL
  ) +
  theme_minimal()
```

Most frequent words (stopwords removed)
Top 15 words by maximum frequency across both texts



Narrative:

Before stopword removal, stopwords completely dominate in frequency across both texts, with similar distributions across them. After stopword removal, topic-specific words are much more visible across the top 15 words, and it is much easier to get a sense for the common substantive material in the texts. For example, both texts clearly relate to “trade,” a high-frequency word across both texts, and they both frequently discuss “money” and “merchants.” At the same time, removing stopwords demonstrates differences in the common words between the text; for example, “exchange,” “Malynes,” and “ballance” are substantive words common in Text A but absent or near-absent from Text B.

There are two clear drawbacks of removing the stopwords that we have removed, because of the information that is made less visible by doing so. First, we have less information about the grammatical and syntactic structure of the texts, because a large portion of the ‘functional’ words that convey this information are stopwords. Second, and relatedly, we have no means for understanding the negation of any of our most common words, because “no” and “not” are stopwords that we have removed.

Section III

```
# Here, I am removing only the tidytext standard
# stopwords, keeping the additional Early Modern
```

```
# stopwords that we previously identified.
words_standard_stopwords_removed <- words |>
  mutate(word = str_to_lower(word)) |>
  anti_join(stop_words, by = "word")

# Here, I am counting the frequency of the words
# and displaying the 15 most-frequent words across
# both texts with only standard tidytext stopwords
# removed.
words_standard_stopwords_removed |>
  count(doc_title, word, sort = TRUE) |>
  pivot_wider(
    names_from = doc_title,
    values_from = n,
    values_fill = 0
  ) |>
  head(15)
```

```
# A tibble: 15 x 3
  word      `Text A` `Text B`
  <chr>      <int>    <int>
1 trade      232      185
2 exchange   186        4
3 fo         184        0
4 haue       163      86
5 malynes    158        0
6 merchants  126      42
7 mony       118        8
8 hee        115      16
9 kingdome   113      75
10 bee       108      51
11 commodities 96      18
12 hath       87      52
13 money       84      81
14 alfo        78        0
15 fuch        69        0
```

Narrative:

Without the custom stopwords removed, “haue,” “bee,” and “hath” have entered the 15 most frequent words, displacing “common,” “balance,” and “becaufe.” These words are simply

archaic spellings/forms of the words “have,” “be,” and “has,” standard functional words that are not specific to the subject matter at hand and convey no information about it.

A research question for which analyzing the frequency of words like these might be valuable could be one where the researcher wants to learn how spellings and word forms have changed over time. Such a project might compare the relative frequency of these words and their modern equivalents in texts written in different years. Stylistic or linguistically focused projects like this demonstrate why there is no universal set of stopwords: stopword lists are meant to remove words that add no “value” to the analysis, but if the words or structures themselves are the subject for analysis, they clearly should not be removed.

Final Reflection

I would tell my naive friend that they are mistaken to believe that the pre-processing decisions we make have no impact on the output of our textual analyses. Clearly, words that are “important” for one project’s purpose (like that of a diachronic, linguistic or stylistic analysis) are removable noise for others (like those focused on the representation of commerce and trade in 16th century England). This exploration has illustrated the evident relevance of stopwords, especially when one is analyzing frequency counts (as stopwords may dominate them before their removal), and also for situating the syntactic and semantic context of the text under review.